Experimental Stress Analysis

UNIT – IV

THEORY OF PHOTOELASTICITY

Course 23A01702c · Professional Elective-IV · JNTUA R23

Syllabus (this unit):

Introduction – Temporary Double Refraction – The Stress-Optic Law – Effects of Stressed Model in a Polariscope for Various Arrangements – Fringe Sharpening – Brewster's Stress-Optic Law.

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1. Introduction to Photoelasticity

- Photoelasticity is a whole-field optical method of stress analysis. A transparent model of a birefringent plastic (epoxy, polycarbonate) is loaded and viewed in polarised light; it shows coloured or dark fringes which give the stress distribution over the entire model.
- It is based on the property of temporary double refraction (birefringence) – a stressed transparent isotropic material behaves like an optical crystal whose axes lie along the principal stress directions.
- Landmarks: 1816 Brewster discovered stress-induced birefringence in glass; 1853 Maxwell gave the stress-optic law; 1930s Coker and Filon developed 2-D photoelasticity; Frocht, Mindlin, Post and Dally – modern developments.

What does photoelasticity give?

- Isochromatic fringes → difference of principal stresses ($\sigma_1 - \sigma_2$) everywhere.
- Isoclinic fringes → directions of the principal stresses everywhere.
- Free boundary: one principal stress is zero, so the boundary stress (where failure usually starts) is obtained directly with high accuracy.

Merits and limitations

- Merits: full-field picture, finds stress concentrations immediately, works for any shape and loading, good for teaching, cheap models for 2-D, extended to 3-D by stress freezing.
- Limitations: needs special model material and a polariscope, model-to-prototype scaling, only elastic stresses; time-edge effect and creep in plastics; separation of σ_1 and σ_2 needs extra work (Unit V); 3-D slicing is laborious.

Types

- 2-D transmission photoelasticity (plane models; Unit V), 3-D photoelasticity (stress freezing, slicing), reflection photoelasticity (a birefringent coating bonded to a real component), scattered-light photoelasticity.

2. Nature of Light and Polarisation

- Light is an electromagnetic transverse wave. The light vector (electric field E) vibrates perpendicular to the direction of travel.

$$c = f\lambda \quad n = c / v \quad E = a \sin[(2\pi/\lambda)(ct - z)] \quad \text{Intensity } I \propto a^2$$

- λ for visible light = 400 – 700 nm; sodium lamp 589 nm; mercury green 546 nm (used in laboratory).
- Refractive index n of a medium = speed of light in vacuum / speed in the medium.

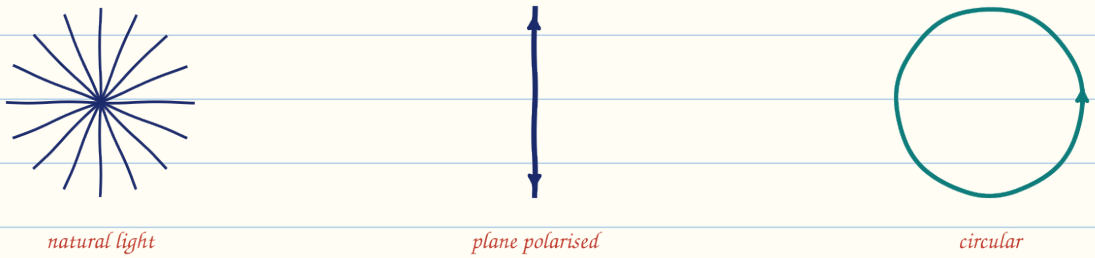


Fig 4.1 Natural, plane-polarised and circularly polarised light

- Natural (unpolarised) light: vibrations in all directions in a plane perpendicular to the ray.
- Plane (linearly) polarised light: vibration only in one plane.
- Circularly polarised light: two perpendicular components of equal amplitude with phase difference $\pi/2$ – tip of E traces a circle; elliptically polarised: unequal amplitude or phase.

Ways of obtaining polarised light

- Reflection at the polarising (Brewster) angle: $\tan\theta_B = n$; absorption (dichroism) in Polaroid sheet – most used in polariscope; double refraction (Nicol prism); scattering.
- Polaroid: a sheet of stretched polyvinyl alcohol with iodine – transmits only the vibration parallel to its transmission axis.

Malus' law

$$I = Io \cos^2\theta \quad (\theta = \text{angle between the polariser and analyser axes})$$

- $\theta = 90°$ (crossed) → $I = 0$ (dark field); $\theta = 0$ (parallel) → $I = Io$ (light field).

3. Double Refraction and Wave Plates

- In anisotropic crystals (calcite, quartz, mica) a ray splits into two:
  - Ordinary ray (O) – obeys Snell's law, speed same in all directions.

- Extraordinary ray (E) – does not obey Snell's law, speed depends on direction.

- Both rays are plane polarised at right angles to each other. The optic axis is the direction along which there is no double refraction.

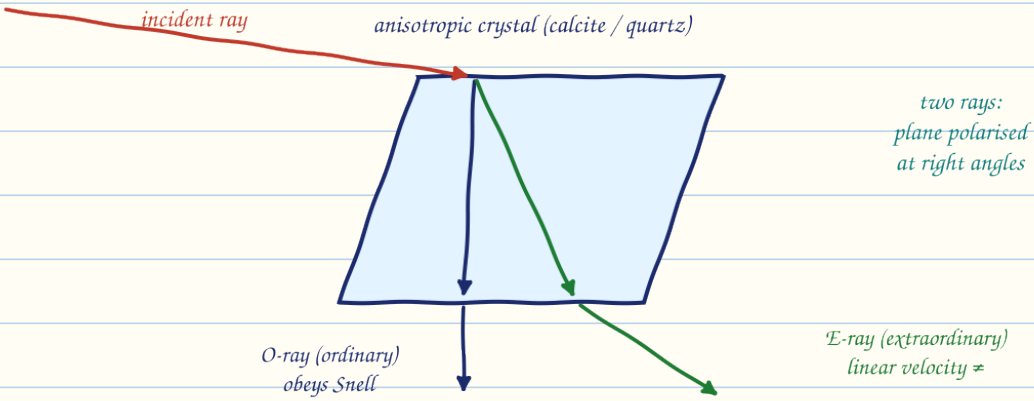


Fig 4.2 Double refraction in a crystal

- In a plate cut parallel to the optic axis the two rays travel along the same path with different speeds and emerge with a relative retardation:

$$\Delta = t\,(n_e - n_o) \text{ phase difference } \delta = 2\pi\Delta/\lambda$$

Wave plates

- Quarter-wave plate (QWP): $\Delta = \lambda/4$ ($\delta = \pi/2$). Plane polarised light at 45° to its axes becomes circularly polarised, and vice versa.
- Half-wave plate: $\Delta = \lambda/2$ ($\delta = \pi$). Rotates the plane of polarisation by 2θ.
- Thickness of a QWP: $t = \lambda/(4\,|n_e - n_o|)$. Quartz ($\Delta n \approx 0.0092$, $\lambda = 589$ nm): $t = 589\times10^{-9}/(4\times0.0092) \approx 16$ µm. Practical QWPs are of mica or plastic film, made for one wavelength.

Circularly polarised light – derivation

- Two components of equal amplitude a along x and y with a phase difference of $\pi/2$ (produced by a quarter-wave plate at 45°):

$$E_x = a\sin\omega t \quad E_y = a\sin(\omega t - \pi/2) = -a\cos\omega t \Rightarrow E_x^2 + E_y^2 = a^2$$

- This is the equation of a circle – the tip of the light vector rotates with constant amplitude; right or left circular depending on the sign of the phase difference.

- Plane polarised light at 45° to the axes of a QWP → circular; circular light through a second QWP (suitably set) → plane polarised again.

4. Temporary Double Refraction

- Many transparent, isotropic, non-crystalline materials (glass, celluloid, Bakelite, epoxy, polycarbonate, gelatin) are optically isotropic when free of stress.
- When stressed, they become doubly refracting like a crystal with optic axes in the directions of principal stresses σ_1 and σ_2; a ray entering the stressed plate splits into two plane polarised components vibrating along σ_1 and σ_2, travelling with different velocities.
- The effect lasts only while the load is applied and disappears on unloading – hence temporary double refraction.
- If the model is heated, loaded and cooled under load, the birefringence is permanently locked (stress freezing – used for 3-D photoelasticity).

Plane vs circular polarisation in the stressed model

- The size of retardation between the two components depends on thickness t and on $(\sigma_1 - \sigma_2)$ – this is the stress-optic law.

5. The Stress-Optic Law (Maxwell) and Brewster's Law

- For a 2-D stress state ($\sigma_3 = 0$) the refractive indices for light vibrating along the principal directions are linear functions of principal stresses:

$$n_1 - n_0 = C_1\sigma_1 + C_2\sigma_2$$
$$n_2 - n_0 = C_1\sigma_2 + C_2\sigma_1$$

- n_0 = refractive index of unstressed material; C_1, C_2 = absolute stress-optic coefficients (properties of material).
- Subtracting:

$$n_1 - n_2 = (C_1 - C_2)(\sigma_1 - \sigma_2) = C\,(\sigma_1 - \sigma_2)$$

- C = relative stress-optic coefficient, unit Brewster: 1 Brewster = 10^{-12} m^2/N.

Brewster's stress-optic law

- Relative retardation (optical path difference) for light passing normally through a model of thickness t:

$$\Delta = (n_1 - n_2)\,t = C\,t\,(\sigma_1 - \sigma_2)$$

- Statement: "The relative retardation developed between the two components is directly proportional to the thickness of the model and to the difference of the principal stresses."
- Number of wavelengths of retardation = fringe order N:

$$N = \Delta/\lambda = C\,t\,(\sigma_1 - \sigma_2)/\lambda \Rightarrow \sigma_1 - \sigma_2 = N\lambda/(C\,t) = N f_\sigma / t$$

- $f_\sigma = \lambda/C$ = material fringe value (N/m per fringe). It depends on material and on wavelength.
- Typical values (λ = 546 nm):
  - Epoxy (Araldite): $C \approx 50$ Brewster, $f_\sigma \approx 11$ kN/m per fringe.
  - Polycarbonate: $C \approx 78$ Brewster, $f_\sigma \approx 7$ kN/m per fringe (very sensitive).
  - PMMA (Perspex): $C \approx 4$–5 Brewster, $f_\sigma \approx 120$ kN/m (insensitive).

Solved examples

- Example 1. Epoxy model, t = 6 mm, f_σ = 11 kN/m. At a point N = 3 fringes.

$$\sigma_1 - \sigma_2 = N f_\sigma / t = 3 \times 11\,000 / 0.006 = 5.5\times10^6\ N/m^2 = 5.5\ MPa$$

- Example 2. Epoxy slab t = 5 mm, C = 50 Brewster, $\sigma_1 - \sigma_2$ = 4 MPa, λ = 546 nm.

$$\Delta = C\,t\,(\sigma_1 - \sigma_2) = 50\times10^{-12} \times 0.005 \times 4\times10^6 = 1.0\times10^{-6}\ m = 1000\ nm\ N = 1000/546 = 1.83$$

- Example 3 (free boundary). At a free edge $\sigma_2 = 0$, so $\sigma_1 = N f_\sigma/t$ directly: for N = 2, f_σ = 7 kN/m (polycarbonate), t = 5 mm → $\sigma_1 = 2 \times 7000/0.005 = 2.8$ MPa.

Remarks

- The law is valid only within the elastic limit and for small, instant loading (creep and the time-edge effect in plastics alter C).
- f_σ depends on wavelength (dispersion) – always state the light used. Use calibration to find f_σ for each model sheet.
- To increase sensitivity choose a material with low f_σ (high C) and a greater thickness.

6. Effect of a Stressed Model in a Polariscope

- Consider a plane polariscope: polariser P and analyser A crossed (axes at 90°), model between them. At a point the principal stress σ_1 makes angle α with the polariser axis.

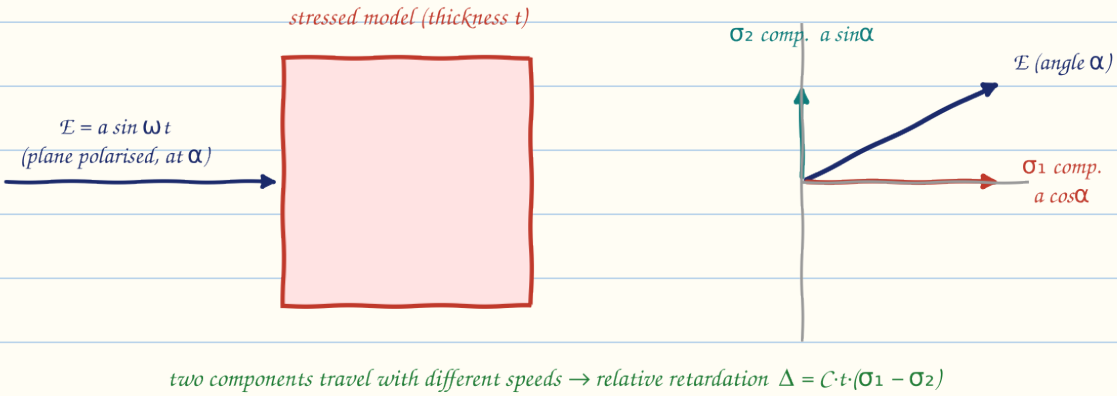


Fig 4.3 Resolution of light vector in a stressed model

- Light emerging from P: $E = a \sin \omega t$. In the model it splits into two components along σ_1 and σ_2:

$$E_1 = a \cos\alpha \sin\omega t \text{ (along } \sigma_1) \quad E_2 = a \sin\alpha \sin\omega t \text{ (along } \sigma_2)$$

- After passing through thickness t, the component with the higher index lags; with relative phase $\delta = 2\pi\Delta/\lambda = 2\pi N$:

$$E_1 = a \cos\alpha \sin\omega t \quad E_2 = a \sin\alpha \sin(\omega t - \delta)$$

- The analyser transmits only the components resolved along its own axis (perpendicular to P):

$$E_A = E_2 \cos\alpha - E_1 \sin\alpha = a \sin\alpha \cos\alpha \,[\sin(\omega t - \delta) - \sin\omega t]$$
$$E_A = -\, a \sin 2\alpha \cdot \sin(\delta/2) \cdot \cos(\omega t - \delta/2)$$

- Amplitude = $a \sin 2\alpha \sin(\delta/2)$, so the intensity of light that reaches the screen:

$$I = K a^2 \sin^2 2\alpha \cdot \sin^2(\pi \Delta/\lambda) = K a^2 \sin^2 2\alpha \cdot \sin^2(\pi N) \text{ (dark-field plane polariscope)}$$

Conditions of extinction (dark points)

- (i) $\sin 2\alpha = 0 \rightarrow \alpha = 0°$ or 90°: the principal stress direction coincides with the polariser (or analyser) axis → black band = isoclinic fringe (independent of

wavelength and of load).

- (ii) $\sin(\pi N) = 0 \rightarrow N = 0, 1, 2, 3, \ldots$: relative retardation is a whole number of wavelengths → dark isochromatic fringe (for monochromatic light).
- Maximum brightness when $\alpha = 45°$ and $N = ½, 3/2, 5/2 \ldots$

Light-field (polariser parallel to analyser)

$$I = K a^2 [\, 1 - \sin^2 2\alpha \cdot \sin^2(\pi N) \,]$$

- Bright where $N = 0, 1, 2, \ldots$; dark where $N = ½, 3/2, \ldots$. So dark-field gives integer orders, light-field gives half orders – both are used for fringe location.

White light

- Each wavelength is extinguished at a different N, so coloured isochromatics appear in the sequence black – grey – white – yellow – red – tint of passage (violet) – blue – green – yellow – red … $N = 0$ is black. Useful to identify the zero-order fringe and the sign of stress; monochromatic light is used for counting.

Sequence of colours with white light (tint of passage)

- Retardation Δ (nm): 0 – black; ≈ 280 – grey; ≈ 450 – white; ≈ 530 – yellow/orange; ≈ 565 – tint of passage (sensitive purple-violet); ≈ 600 – blue; ≈ 700 – blue-green; then the second-order series repeats with brighter colours.
- The tint of passage (retardation ≈ 1 fringe ≈ 575 nm) changes abruptly from red to blue – a sharp reference for fringe order 1. Beyond about the 3rd–4th order the colours wash out to pink-green.

7. Polariscope Arrangements

7.1 Plane (linear) polariscope

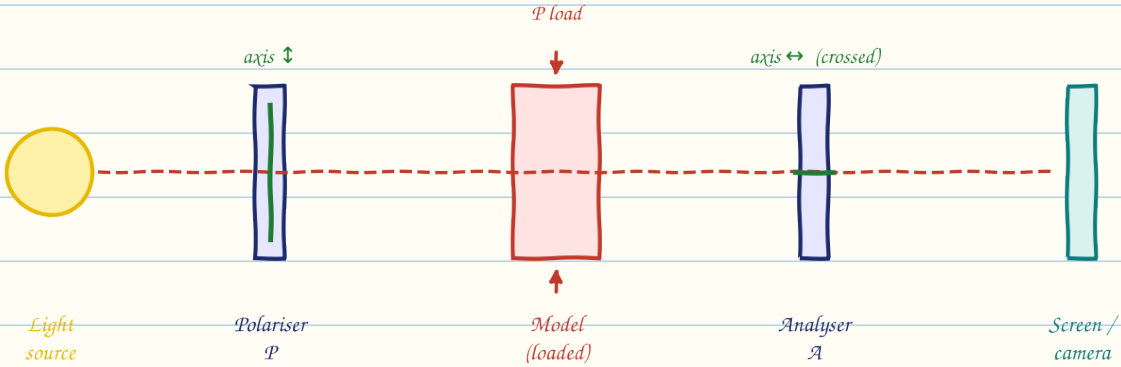


Fig 4.4 Plane polariscope (crossed arrangement)

- Components: light source (white or monochromatic), polariser P, loaded model M, analyser A, screen / camera.
- Crossed (dark field): axes of P and A perpendicular – $I = K a^2 \sin^2 2\alpha \sin^2 \pi N$; dark field when no model or no stress.
- Parallel (light field): axes parallel – bright field.
- A plane polariscope shows both isochromatics and isoclinics together (isoclinic black band confuses the picture).

7.2 Circular polariscope (adds two quarter-wave plates)

- Order: source → P → Q1 (QWP at 45° to P) → model → Q2 (QWP) → A → screen. Q1 converts plane polarised light into circularly polarised light, so the isoclinic dependence on α disappears and only isochromatics are visible.
- Four standard arrangements (P, A: polariser and analyser; Q1, Q2: quarter-wave plates):
- A: P ⊥ A, Q1 ⊥ Q2 (crossed) – dark field: $I = K a^2 \sin^2(\pi N)$
- B: P ⊥ A, Q1 ∥ Q2 (parallel) – light field: $I = K a^2 \cos^2(\pi N)$
- C: P ∥ A, Q1 ⊥ Q2 – light field: $I = K a^2 \cos^2(\pi N)$
- D: P ∥ A, Q1 ∥ Q2 – dark field: $I = K a^2 \sin^2(\pi N)$
- Procedure to choose arrangement: dark field (A or D) gives integer fringe orders ($N = 0, 1, 2 \ldots$); light field (B or C) gives half orders ($N = ½, 3/2 \ldots$). Combined use improves location of fringe centres.

Plane polariscope or circular polariscope?

- Isoclinics: use plane polariscope (rotate P and A together). Isochromatics: use circular polariscope. (Unit V)

Components of a practical polariscope

- Light source (high-pressure mercury vapour lamp with green filter 546 nm, or sodium lamp 589 nm; white tungsten lamp for colours) + diffuser.
- Field (collimating) lenses – produce a parallel beam; diameter decides the size of the model that can be viewed (e.g. 150 – 300 mm).
- Polariser, analyser and two quarter-wave plates in rotating mounts with graduated circles (to set angles).
- Loading frame (screw or dead-weight) with a load cell; model holder.
- Camera lens / screen / digital camera to record the fringe pattern.

Plane versus circular polariscope

- Plane: only P and A; shows isochromatics + isoclinics; used to find principal stress directions.
- Circular: P, Q1, Q2, A; shows only isochromatics; used to find $\sigma_1 - \sigma_2$.
- Plane polariscope needs no wave plates, so no wavelength error; the circular one is exactly correct only for the wavelength for which the QWPs were made.

Stress freezing (3-D photoelasticity) in brief

- Epoxy model heated slowly to the critical (stress-freezing) temperature (~ 120 °C) → loaded → cooled very slowly (2–3 °C/h) under load → load removed.
- The birefringence stays locked; the frozen fringe value is only about 0.3 – 0.5 kN/m (about 20–40 times more sensitive than at room temperature).
- The model is sliced (3 – 6 mm thick slices) and each slice is examined in a polariscope like a 2-D model.

8. Fringe Sharpening (and Multiplication)

- Problem: the intensity law $I \propto \sin^2(\pi N)$ is a smooth cosine curve, so the dark fringes are broad and their centres are difficult to locate, especially at fractional orders and when fringes crowd together.
- Fringe sharpening = reducing the width of the fringes so that the positions of integer (or half) orders can be located accurately. Fringe multiplication =

increasing the number of fringes in the model so that regions of low stress show measurable fringes.

Methods

- Monochromatic light (sodium or mercury) – black, well defined fringes instead of coloured bands.
- Post's method of fringe sharpening and multiplication – a series of partial mirrors (partially reflecting plates) placed before and after the model (parallel to it) at the polariser/analyser gives multiple passes of the light through the model. Intensity pattern becomes

  $I \propto \sin^2(\pi N) / [\, 1 + F \sin^2(\pi N)\,]$ (F depends on mirror reflectivity)

  - The result is very narrow bright (or dark) fringes and multiplied fringes ($N \to 2N, 3N, \ldots$) – analogous to the sharp fringes of a Fabry–Perot interferometer.
- Photographic methods – high-contrast (high gamma) film, double exposure, reducing background.
- Digital image processing – video camera, threshold and skeletonise (thin) the fringes; computer locates the fringe centres.
- Compensation techniques (Tardy, Babinet-Soleil) – measure fractional fringe order without sharpening (Unit V).

Why it matters

- Accurate fractional fringe order at the point of interest → accurate stress; essential at stress concentration (fillets and holes) where fringes crowd.

9. Quick Revision – Unit IV

- Photoelasticity = temporary double refraction + polarised light → whole-field stresses.

Stress-optic law: $n_1 - n_2 = C(\sigma_1 - \sigma_2)$; $\Delta = C\,t\,(\sigma_1 - \sigma_2)$; $\sigma_1 - \sigma_2 = N f_\sigma / t$; $f_\sigma = \lambda / C$

Plane polariscope: $I = K a^2 \sin^2 2\alpha \sin^2 \pi N$ (isoclinics from $\sin 2\alpha$, isochromatics from $\sin \pi N$)

- Circular polariscope: 4 arrangements, $I = K a^2 \sin^2 \pi N$ (dark) or $\cos^2 \pi N$ (light) – no isoclinics.
- Fringe sharpening: monochromatic light, Post's partial mirrors, digital processing.
- Units: Brewster = 10^{-12} m^2/N; fringe value in N/m per fringe.

Important questions

- What is temporary double refraction? Explain with a neat sketch.
- State and derive the stress-optic law. Define Brewster, fringe value and fringe order.
- Derive the expression for the intensity of light emerging from an analyser in a plane polariscope with a stressed model. Explain isoclinics and isochromatics.
- Describe the arrangements of a circular polariscope with intensity equations.
- What is fringe sharpening? Explain Post's method.
- A model of thickness 6 mm with f_σ = 11 kN/m shows N = 4; find $\sigma_1 - \sigma_2$. (Ans: 7.33 MPa)